

Circulation Amendment — Module 5: Seam Ledger

PLOL-conformant brief section for the v0.13 packet. Prepared 2026-07-14. Companion to `SEAM_TO_LEDGER_MAP.md` (the design record) and `AHCKernel/SeamLedger.lean` (the machine-checked object). Structure follows PLOL: **(I) Plain-Language Summary · (II) Formal Appendix · (III) Theorem-to-Text Mapping**. The v0.12 circulation brief is retained unchanged as the artifact of its round; this amendment is the Module 5 addendum, not a rewrite of that brief.

I. Plain-Language Summary

What was already true. Through v0.12 the kernel is strong against *malformed* inputs: you cannot build an under-provisioned sensor network, a certificate that authorizes an unsafe action, or an emergency day during a Structural Review. Those are theorems about badly-formed objects.

The gap Module 5 addresses. The kernel had no defense against a *well-formed lie from a legitimate authority*. Its most load-bearing inputs — the request to open an emergency, the signal that a threat was confirmed, the claim that a matter is "materially new," the danger threshold, the certificate that an action is reversible, the mark that a finding is or is not "critical," and the Layer 0 dispositions that close a review — were **naked values**: a single actor could assert any of them, and the kernel would reason correctly from a false premise. The strongest theorems in the system rest on inputs nobody was required to justify.

What Module 5 does. It wraps each such input in a `SeamClaim`: a **dated, attributed, contestable record** carrying who authorized the act, what evidence supports it, how that evidence is held in custody, what warnings were available, what counter-evidence exists, what observation would prove the claim wrong, and — critically — the affected population's own counter-record. Building the wrapper for an emergency now costs a coordinated, attributable, auditable act by named roles, not the flip of a bit.

What Module 5 does NOT do — stated plainly and repeated everywhere. It **does not catch lies**. It has no `truth` field and proves no payload true. A false emergency can still be declared; a false certificate can still be issued; an official can still under-mark a finding. What changes is that each of these becomes a **visible multi-party act with preserved contradictions** instead of a silent administrative utterance. The system converts lying from one person's cheap assertion into a documentary event that names its participants and carries its own refutation. That, and only that, is the claim.

The one guarantee that is a hard wall, not a cost. A community filing under the ledger can *preserve, disclose, demand audit, and demand provisional restraint* — and it can do **none** of these things become punishment. The two are different types with no function between them (`step_one_never_implies_step_two`). This is deliberate: it is what keeps the ledger from becoming a new unconstrained authority. An allegation of discernment-corruption never, by itself, unlocks coercion.

II. Formal Appendix

Module 5 is `AHCKernel/SeamLedger.lean`: core-only Lean 4.15.0, no Mathlib, 15 audited theorems, axiom footprint at most `[propext, Quot.sound]` (11 of the 15 axiom-free), zero `sorry`. It imports Modules 1 and 2 to bind its theorems to the real episode machine and certificate layer.

The wrapper. `SeamClaim` α carries payload : α (the naked v0.12 value verbatim) plus `incidentId`, `issuer`, `authorizationChain`, `evidenceRefs`, `sourceBundleDigest`, `custodyStatus` (the ledger's five-rung custody ladder `inLedger` $\rightarrow$ `captureRequired` $\rightarrow$ `locatorVerified` $\rightarrow$ `hashedPendingBackup` $\rightarrow$ `verified`), `warningsAvailable`, `counterEvidence`, `falsificationPath`, `affectedCounterRecord`, `conflicts`, and a procedural status. It has **no truth field**, by design.

Tier A — state-machine invariants.

Thm	Name	Statement (informal)
L1	<code>no_naked_authority_bit</code>	An <code>issue</code> accompanied by no authorization chain or no evidence is never issued (the gate returns <code>false</code>).
L1'	<code>naked_issue_no_pending</code>	Bound to <code>estep</code> : from <code>idle</code> , a gate-failing <code>SeamClaim</code> produces no pending PIO.
L2	<code>counterrecord_persists(_run)</code>	No official transition (<code>setStatus</code> / <code>addEvidence</code> / <code>reclassifyConflicts</code> / <code>close</code>), singly or in any finite sequence, alters an attached counter-record.
L2'	<code>counterrecord_digest_stable</code>	The counter-record stays bound to the same event hash across every official transition.
L3	<code>step_one_never_implies_step_two</code>	No act a community/ledger trigger makes available (<code>preserve</code> / <code>disclose</code> / <code>audit</code> / <code>provisionalRestraint</code>) is a step-two (<code>punish</code> / <code>irreversibleEnforce</code>) authorization.
L4	<code>remedy_label_not_remedy</code>	A remediation in stage <code>planIssued</code> does not close a Structural Review; <code>remedy_stages_distinct</code> proves the lifecycle stages are distinct by construction.
L5	<code>contested_cannot_unlock_irreversible</code>	A <code>SeamClaim</code> with <code>status = materiallyContested</code> cannot satisfy the unlock predicate for any action.
L5'	<code>seam_trace_stays_inside</code>	Composed with W21: an uncontested authorized sub-causal trace from inside the certified region ends inside it; <code>contested_blocks_trace</code> refuses a contested trace.

Tier B — opaque-predicate relations. Each relation the kernel cannot verify is modelled as an abstract predicate the theorem quantifies over every valuation of — the pattern Module 3 already uses for sensor honesty (ground truth unavailable to the system). The kernel proves the structural property for *all* valuations; it never decides the oracle.

Thm	Name	Structural property proved	What is NOT proved (the oracle)
L6	<code>issuer_not_sole_validator</code>	A multi-party validation has a validator distinct from the issuer.	Whether a listed actor truly validated (M5-O1).
L7	<code>interested_actor_cannot_self_close</code>	For every interest relation: if all validators are interested, the close is not accepted.	Who is "interested" (M5-O5).
L8	<code>challenge_requires_merits_response</code>	A rebuttal is unconstructible without an evidence reference; an undisposed challenge is unresolved.	Whether the rebuttal is correct (M5-O9).
L9	<code>criticality_cannot_suppress_typed_conflict</code>	An affected-population counter-record bound to an event's digest survives	Who may <i>weight</i> the counter-record (M5-O8,

Thm	Name	Structural property proved	What is NOT proved (the oracle)
		official re-classification.	deferred).

Modeling disclosures (labelled contestation targets). The wrapper **relocates** trust boundaries; it does not eliminate them. The nine new oracles M5-O1..O9 are enumerated in `SEAM_TO_LEDGER_MAP.md §3`. Three carry the boundary constraint that **no LLM classifier decides them** (M5-O3, M5-O5, M5-O7, M5-O9 — all protected-path judgments). One (M5-O8, weighted community standing) is **deferred pending governance machinery** for representation, revocation, and intra-population disagreement — it is *not* stubbed with a Boolean; the counter-register currently preserves any attributed attachment, and standing to weight it is future work. If a reviewer finds a tenth oracle the module hides, that is a finding against the map.

Deprecation. v0.13 ships the wrappers additively; the naked `EpInput / Envelope / Event` interfaces are marked deprecated with a removal target of **v0.14**. A preserved bypass is technical debt made normative; retaining the naked interfaces this round keeps the 116 v0.12 theorems stable, and the gating migration is scheduled, not indefinite.

CI expectations (updated). `docs/MANIFEST_v0.13.txt`. The build asserts **131 audited theorems, 60 axiom-free** (v0.12's 116/49 plus Module 5's 15/11), every theorem at most `[pronext, Quot.sound]`, never `Classical.choice`, zero `sorry`. Source digest `eca4b27e24d0999b90d913857839b4806b8ea7764b7ba0aa1f7c84bd9b1940ea`.

III. Theorem-to-Text Mapping

Module 5 is the first module whose source text is **not** AHC v3.1 / Companion A but the **Veriticide General Ledger** — cited as origin, not merged. The mapping rows are therefore cross-repository citations. Full quotation-level rows are appended to `docs/normative/THEOREM_TO_TEXT_MAP.md` under "Module 5"; the correspondence in brief:

Kernel object	Veriticide source (machinery, not corpus)
<code>SeamClaim</code> fields (chain / evidence / warnings / counter-evidence / custody / falsifier / counter-record)	Documentation & Standing Protocol v0.1 §2 (six tracks A–F); §3 (ADVERSARIAL CHECK, COUNTER-EVIDENCE STATUS); case-file grammar <code>00-charge-theory ... 05-custody-manifest</code>
<code>CustodyStatus</code> ladder	<code>cases/*/05-custody-manifest.md §1</code> (the five custody states)
<code>FalsificationCondition</code>	<code>cases/*/04-falsification-memo.md</code>
L3 <code>step_one_never_implies_step_two</code>	Standing Protocol §7 (the two-step position; "the clause that does not narrow away"); §7-bis (Forum-Now toggle)
L4 <code>remedy_label_not_remedy</code>	Convention/Protocol remedy-vs-label distinction; the toggle's "flipping is a logged act"
L8 <code>challenge_requires_merits_response</code>	Standing Protocol §3 (the two mandatory analytical fields)
L9 <code>criticality_cannot_suppress_typed_conflict</code>	Provenance Protocol §3 counter-register; Module 4 P4's disclosed incompleteness turned to the community's benefit
Cost-to-fake analyses (map column d)	Provenance Grading & Corpus-Absorption Protocol, Part II §5 (the cost-to-fake rule)

The honest deviation a faithfulness reviewer should examine first: unlike Modules 1–4, whose theorems formalize sentences of a fixed constitutional text, Module 5's theorems formalize the *procedural machinery* of a separate working repository. The

correspondence is structural (a `SeamClaim` field $\leftrightarrow$ a protocol track), not quotational (a theorem $\leftrightarrow$ a v3.1 sentence). This is stated so it is not mistaken for the tighter Module 1–4 mapping.